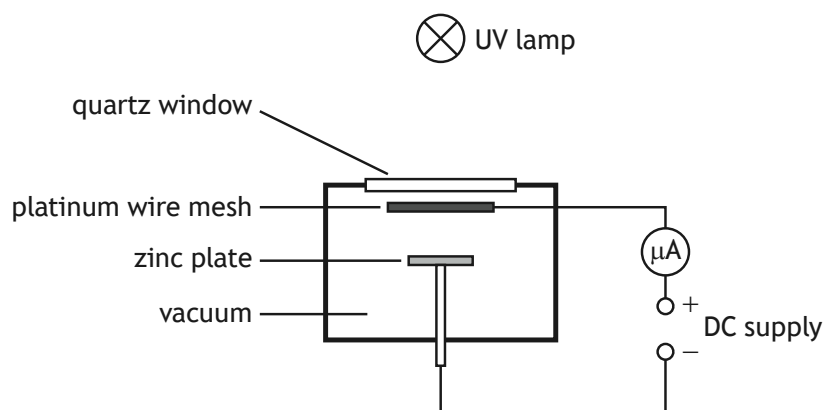


9. A student uses the apparatus shown to investigate the photoelectric effect.



When the lamp is switched on, ultraviolet (UV) radiation passes through the quartz window and wire mesh, and is incident on the zinc plate.

A current of $5.00 \mu\text{A}$ is produced in the circuit.

- (a) (i) The lamp emits UV radiation with a frequency of $1.25 \times 10^{15} \text{ Hz}$.

Calculate the energy of a photon of the UV radiation.

Space for working and answer

3

- (ii) The work function of zinc is $5.81 \times 10^{-19} \text{ J}$.

Explain why UV radiation with a frequency of $1.25 \times 10^{15} \text{ Hz}$ produces a current in this circuit.

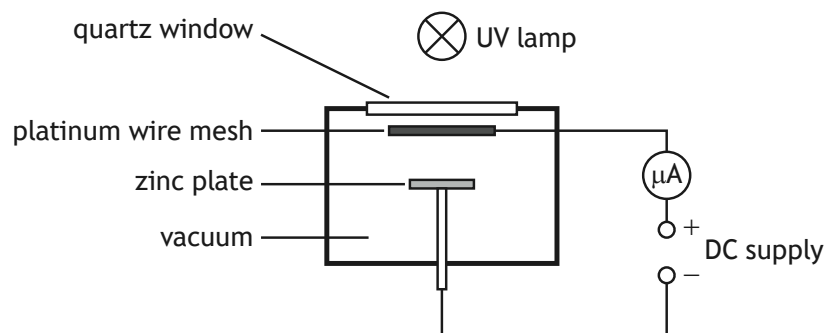
2



* X 8 5 7 7 6 0 1 3 0 *

9. (continued)

(b) The student now moves the UV lamp closer to the quartz window.



State whether the current in the circuit increases, decreases, or stays the same.

You must justify your answer.

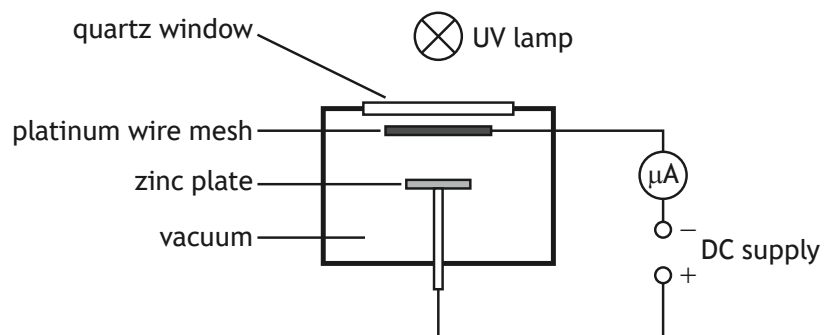
2

[Turn over



9. (continued)

(c) The student now reverses the polarity of the DC supply.



State the effect this change has on the current in the circuit.

Justify your answer.

2

